

Introduction to Statistical Methods
(S2-25 AIMLCLZC418) – Assignment 2

AIML -Section 2

Each question carries 5 marks .

Total - 10 Marks

Duration: 25th July 2026 – 8th August 2026

- 1) Submissions are individual**
- 2) Solve these on paper, scan, and upload**
- 3) Plagiarism results in zero marks**
- 4) Write your name, BITS ID and Section on each page**
- 5) Only handwritten solutions with formula, full steps with proper justification are required.**

1. An investigator wants to examine the hypothesis that husbands are more intelligent than their wives. IQ scores of 10 couples are recorded as follows.

Husband's IQ	117	105	97	105	123	109	86	78	103	107
Wife's IQ	106	98	87	104	116	95	90	69	108	85

Test the hypothesis with a reasonable test at the 5% level of significance. [5 Marks]

2. Teaching Method vs Subject (Education)

A university wants to study whether students' performance depends on the subject and the teaching method. Three teaching methods are applied to four subjects, and the average marks obtained are shown below.

Subject	Method A	Method B	Method C
Mathematics	78	82	85
Physics	74	79	81
Chemistry	80	84	86
Programming	85	88	91

Question: [5 Marks]

- a) Identify the two factors involved in the experiment.
- b) State the null hypotheses for both factors.
- c) Perform a two-way ANOVA (without replication) at the 5% level of significance.
- d) Interpret whether teaching method and subject significantly affect student performance.

----ALL THE BEST----